
(a) TX eye through the channel
PRBS7 bits
PWL file or
XSPICE source
DUT: 2:1 muxes
+ 3-tap driver
(extracted)
pad + ESD
$C_p \approx 0.6\,\mathrm{pF}$
channel model
RLGC ladder / LTRA,
or $h(t)$ as FIR
100 Ω load
+ 300 fF
wrdata →
Python eye,
eye_scan
ideal 2.5 GHz
clock (or PLL)
pass: eye ≥ model eye −10%
at all corners
(b) Sampler offset, noise and metastability
V_{in} steps
$-5 \ldots +5\,\mathrm{mV}$
around V_{CM}
DUT: CML sampler
+ offset DAC
Monte Carlo mismatch
count 1s over
N clocks →
$P(1)$ vs V_{in}
fit $P(1) = \Phi\left(\frac{V_{\mathrm{in}} - V_{\mathrm{os}}}{\sigma}\right)$
→ V_{os} histogram, σ
tiny V_{in} sweep:
$t_{\mathrm{CK}\to\mathrm{Q}}$ vs $\ln V_{\mathrm{in}}$
slope → τ
2.5 GHz clock
(transient noise on)
pass: $\sigma \le 1\,\mathrm{mV}$, $|V_{\mathrm{os}}| \le 0.5\,\mathrm{mV}$ after trim, $\tau \le 5\,\mathrm{ps}$